

Demonstration of Proposed Features

Date	3 July 2026
Team ID	SWTID-2026-9706
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Audio Upload & Playback	User uploads WAV/MP3 and plays it back in-browser	Implemented	Yes	Verified working in live app
2	Speech-to-Text Transcription	Converts uploaded audio into text using OpenAI Whisper	Implemented	Yes	Tested and confirmed accurate transcription
3	Semantic Similarity Scoring	Compares transcript to reference concept using Sentence-BERT	Implemented	Yes	Verified against known worked example (score matched exactly)
4	Audio Feature Extraction	Extracts pause ratio and RMS energy using Librosa	Implemented	Yes	Timing and output verified directly
5	Filler Word Detection	Calculates filler word ratio from transcript	Implemented	Yes	Confirmed correct calculation
6	Scoring Engine	Combines all metrics into final score & understanding level	Implemented	Yes	Formula verified to exactly match reference example (60/100 → Moderate)
7	Waveform Visualization	Generates and displays waveform plot	Implemented	Yes	Confirmed rendering correctly in UI and PDF
8	PDF Report Generation	Downloadable PDF with transcript, waveform, evaluation summary	Implemented	Yes	Tested end-to-end, PDF generated and verified

Feature Implementation Summary:

Total Features Proposed	8
Total Features Implemented	8
Total Features Demonstrated	8
Overall Implementation Rate (%)	100%